Task

Use the answers to complete the Sudoku grid. Answers of one digit are unshaded, answers with two digits are grey and answers with three digits are green.

Remember:

- One digit per square
- Every column, row and 3x3 box must contain the numbers 1-9.

9	*e*	1	3	4	*n*	8	6	*y*
j	*s*	7	8	*m*		9		*c*
6		8	7	*o*		1	*t*	4
5	7	2	9	3	4	6	1	*p*
*a**		4			*r*	*g*	9	*x*
a	9	*l*		7	*d*	4	5	*f*
7	*v*	5	*w*	9	3	2	8	
k		*h*	1	*b*	7	*u*	4	9
4	*i*	9	*z*	6	*q*		7	1

Clues

a*. The relative atomic mass of oxygen
a. The number of protons in an oxygen atom
b. How many atoms are in $CaCO_3$
c. How many atoms are in Al_2O_3
d. What is the relative mass of a neutron?
e. The number of hydrogen atoms in a water molecule
f. The number of electrons in an atom of iron
g. Which group of the periodic table contains elements known as the halogens?
h. How many hydrogen atoms are in ethanol: CH_3CH_2OH?
i. Which group in the periodic table contains only 1 non-metal element; all the rest are metals?
j. How many elements are in H_2SO_4?
k. What is the maximum number of electrons in the first shell of an atom?
l. How many protons are in a krypton atom?
m. The relative mass of a proton
n. What is the total relative formula mass of CaO?
o. How many electrons are in a copper atom?
p. The maximum number of electrons in the second shell of an atom
q. What is the atomic number of astatine?
r. What is the electron configuration of phosphorous?
s. How many neutrons are in a bromine atom?
t. What is the relative atomic mass of sulfur?
u. How many electrons are in a lithium atom?
v. What is the total number of electrons in HCl?
w. How many electrons are in the outer shell of a carbon atom?
x. How many oxygen atoms are in sulphur trioxide?
y. How many neutrons are in a nitrogen atom?
z. How many electrons are in the outer shell of a

	magnesium atom?

Answers

9	2	1	3	4	5	8	6	7
3	4	7	8	1	6	9	2	5
6	5	8	7	2	9	1	3	4
5	7	2	9	3	4	6	1	8
1	6	4	5	8	2	7	9	3
8	9	3	6	7	1	4	5	2
7	1	5	4	9	3	2	8	6
2	8	6	1	5	7	3	4	9
4	3	9	2	6	8	5	7	1

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